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This handout is designed to help writers develop and use logical arguments in writing. This handout helps writers analyze the arguments of others and generate their own arguments. However, it is important to remember that logic is only one aspect of a successful argument. *Non-logical arguments*, statements that cannot be

logically proven or disproved, are important in argumentative writing—such as appeals to emotions or values. *Illogical arguments*, on the other hand, are false and must be avoided.

Logic is a formal system of analysis that helps writers invent, demonstrate, and prove arguments. It works by testing propositions against one another to determine their accuracy. People often think they are using logic when they avoid emotion or make arguments based on their common sense, such as "Everyone should look out for their own self-interests" or "People have the right to be free." However, unemotional or common sense statements are not always equivalent to logical statements. To be logical, a proposition must be tested within a logical sequence.

The most famous logical sequence, called **the syllogism**, was developed by the Greek philosopher Aristotle. His most famous syllogism is:

Premise 1: All men are mortal.

Premise 2: Socrates is a man.

Conclusion: Therefore, Socrates is mortal.

In this sequence, premise 2 is tested against premise 1 to reach the logical conclusion. Within this system, if both premises are considered valid, there is no other logical conclusion than determining that Socrates is a mortal.

This guide provides some vocabulary and strategies for determining logical conclusions.